

Relevance Model: Math Spec

allocation_agent.ml

We learn $p_\theta(y = 1 \mid \text{job}) \in [0, 1]$, the probability a JOBCANDIDATE reaches mid-stage (callback / screener / offer) versus rejection / no-response.

1 Featurization

A job j becomes a fixed-length vector $\mathbf{x}(j) \in \mathbb{R}^d$ with $d = 64 + 7 + 4 = 75$, the concatenation of three blocks.

Title (hashed bag-of-words, 64 dims). Tokenize the title with the regex `[A-Za-z0-9]+`, lowercase to get $T(j) = (t_1, \dots, t_m)$. With $h : \Sigma^* \rightarrow \{0, \dots, 63\}$ a fixed BLAKE2B-derived hash, define raw counts

$$c_k(j) = \sum_{i=1}^m \mathbf{1}[h(t_i) = k], \quad k = 0, \dots, 63,$$

and L_2 -normalize to remove length effects:

$$\mathbf{x}_k^{\text{title}}(j) = \frac{c_k(j)}{\max\left(1, \sqrt{\sum_{\ell} c_{\ell}(j)^2}\right)}.$$

ATS (one-hot, 7 dims). With vocabulary

$\mathcal{A} = (\text{greenhouse, lever, ashby, workday, workable, icims, unknown})$

and any out-of-vocabulary value mapped to *unknown*,

$$\mathbf{x}_a^{\text{ats}}(j) = \mathbf{1}[\mathcal{A}_a = \text{ats}(j)], \quad a = 0, \dots, 6.$$

Numeric (4 dims). Let $\Delta(j) = \max(0, t_{\text{now}} - t_{\text{posted}}(j))$ in seconds and $\Delta_d(j) = \Delta(j)/86400$ in days. Each entry is squashed to roughly $[0, 1]$:

$$\mathbf{x}^{\text{num}}(j) = \begin{pmatrix} \min(|\text{title}(j)|, 200) / 200 \\ \min(\Delta_d(j), 365) / 365 \\ \text{clip}(p_{\text{cb}}(j), 0, 1) \\ \min(\log(1 + \Delta_d(j)) / \log(1 + 365), 1.5) \end{pmatrix},$$

where $p_{\text{cb}}(j)$ is the upstream prior `expected_callback_prob`.

Final vector. $\mathbf{x}(j) = \mathbf{x}^{\text{title}}(j) \parallel \mathbf{x}^{\text{ats}}(j) \parallel \mathbf{x}^{\text{num}}(j) \in \mathbb{R}^{75}$.

2 Labels

The mapping $\ell(\cdot) \in \{0, 1\}$ collapses heterogeneous status strings:

$$\ell(s) = \begin{cases} 1 & s \in \{\text{callback, screener, offer}\} \text{ or } s \in \{1, \text{true, yes}\}, \\ 0 & s \in \{\text{rejection, no_response, none}\} \text{ or } s \in \{0, \text{false, no}\}. \end{cases}$$

3 Model

A two-hidden-layer MLP with ReLU and dropout produces a single logit $z_\theta(\mathbf{x}) \in \mathbb{R}$:

$$\begin{aligned} \mathbf{h}_1 &= \text{Dropout}_p(\text{ReLU}(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1)), & \mathbf{W}_1 &\in \mathbb{R}^{H \times d}, \\ \mathbf{h}_2 &= \text{Dropout}_p(\text{ReLU}(\mathbf{W}_2 \mathbf{h}_1 + \mathbf{b}_2)), & \mathbf{W}_2 &\in \mathbb{R}^{H \times H}, \\ z_\theta(\mathbf{x}) &= \mathbf{w}_3^\top \mathbf{h}_2 + b_3, & \mathbf{w}_3 &\in \mathbb{R}^H, \end{aligned}$$

with defaults $d=75$, $H=64$, $p=0.1$. The probability head is $p_\theta(y=1 \mid \mathbf{x}) = \sigma(z_\theta(\mathbf{x}))$ where $\sigma(z) = (1 + e^{-z})^{-1}$.

4 Loss

Class-weighted binary cross-entropy on logits, with positive-class weight

$$w_+ = \begin{cases} |\mathcal{N}|/|\mathcal{P}| & |\mathcal{P}|, |\mathcal{N}| > 0, \\ 1 & \text{otherwise.} \end{cases}$$

For a batch $\{(\mathbf{x}_i, y_i)\}_{i=1}^N$,

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N \left[w_+ y_i \log(1 + e^{-z_i}) + (1 - y_i) \log(1 + e^{z_i}) \right], \quad z_i := z_\theta(\mathbf{x}_i).$$

This is the BCE form used by `BCEWithLogitsLoss(pos_weight=w+)` and is numerically stable via the log-sum-exp trick.

5 Optimization

Full-batch Adam with L_2 weight decay λ :

$$\theta \leftarrow \text{Adam}(\theta, \nabla_\theta \mathcal{L}(\theta) + \lambda \theta),$$

defaults $\eta = 10^{-2}$, $\lambda = 10^{-4}$, 200 epochs. Splits use a seeded shuffle with validation fraction 0.2 (at least one row) and degrade to no validation if the corpus is too small to spare.

6 Evaluation

Threshold accuracy at $\tau = 0.5$:

$$\text{Acc} = \frac{1}{N} \sum_i \mathbf{1}[\mathbf{1}[\hat{p}_i \geq \tau] = y_i].$$

AUC via the Mann–Whitney U formulation, which is robust to ties:

$$\text{AUC} = \frac{1}{|\mathcal{P}||\mathcal{N}|} \sum_{i \in \mathcal{P}} \sum_{j \in \mathcal{N}} \left(\mathbf{1}[\hat{p}_i > \hat{p}_j] + \frac{1}{2} \mathbf{1}[\hat{p}_i = \hat{p}_j] \right).$$

If either class is empty in the validation split, AUC is reported as 0.5 (no signal).

7 Inference

At score time the same $\mathbf{x}(j)$ is built (deterministic hashing keeps train / serve parity) and the service returns $\hat{p}(j) = \sigma(z_\theta(\mathbf{x}(j)))$ per job.